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Physical activity interventions in pediatric, adolescent and young adult cancer survivors: A systematic review

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Abstract

Purpose: The aim was to summarize the current literature for the effectiveness of physical activity interventions on physical functioning, body composition, and quality of life (QOL) in adolescent and young adult cancer survivors.

Methods: We conducted systematic structured searches of PubMed and Web of Science databases. Two independent researchers selected against inclusion criteria: (1) lifestyle intervention including physical activity and/or physical activity interventions for pediatric, adolescent, and young adults with any cancer diagnosis; (2) measured QOL, physical functioning (ex. strength, activities of daily living), or body composition (ex. Changes in weight, percent body fat); and (3) randomized controlled trials.

Results: Searches identified 4,770 studies. Following the removal of duplicates and title and abstract screening, 83 full-text articles were assessed, and 9 studies met the inclusion criteria. Childhood and adolescent cancer survivors encompassed n=7 studies while young adult cancer survivors were included in n=2 studies. Three studies reported using a theoretical framework and six did not. Interventions ranged from one week to six months. Across all studies reviewed, n=2 reported improvements in physical activity, n=5 studies reported partial improvements, and n=2 reported no improvements.

Conclusions: Interventions to improve physical activity behaviors reported mixed results. Only two physical activity interventions incorporated young adults with cancer, thus physical activity

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The authors declare that they have no competing interests.

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interventions for young adult cancer survivors should be further explored. Future research should focus on personalized physical activity components encouraging behavior change techniques to maximize physical health and QOL improvements.

Keywords

survivorship; physical activity; health outcomes; childhood; young adult; quality of life

Introduction:

Advancements in treatments and care have greatly improved the survival rates of childhood and young adult cancer survivors, resulting in a growing number of people living beyond the disease [1-3]. However, the completion of cancer treatment does not represent an endpoint for many childhood and young adult cancer survivors as the risk of chronic diseases is estimated to be 10 times greater than their siblings' risk [4] including comorbid conditions such as type II diabetes, osteoporosis, and cardiovascular disease [2, 5-7]. Roughly 66-95% of childhood and young adult cancer survivors experience health problems that occur months or years following treatment completion, known as late effects, [5, 8-10], and 80% of these patients require additional medical attention resulting from late effects [11].

A mere 35% of childhood and young adult cancer survivors report being aware that a history of cancer treatment may lead to future health problems [7, 12]. Therefore, childhood and young adult cancer survivors should be educated on health risks associated with cancer and treatment [7], with the goal of decreasing comorbidities and adverse events (e.g. mortality, morbidity, and functional impairment) by increasing health-promoting behaviors. The American Cancer Society (ACS) and American Institute for Cancer Research (AICR) guidelines include weekly physical activity recommendations for cancer survivors including 150 to 300 minutes of moderate intensity activity per week (ACS) and 30 minutes of moderate-vigorous activity 5 times a week (AICR), yet nearly 25% of childhood and young adult survivors are completely sedentary [13-15]. High rates of sedentary behavior in childhood and young adult survivors may result in worse quality of life (QOL) and greater symptomatology, including fatigue [16], constipation [17], frailty [18], low muscle mass [18], slow walking speed [18], and weakness similar to that of adults aged 65 and older [18]. Healthy lifestyle interventions for cancer survivors that incorporate regular physical activity may reduce the risk of late effects and comorbidities [19]. In the general non-cancer population, physical exercise programs have been shown to improve physical and psychological QOL; increase aerobic fitness, muscle strength, and bone mass; and facilitate weight loss in both adults and children [20-23].

The purpose of this study was to conduct a systematic review of the literature pertaining to the prevalence of physical activity interventions in childhood and young adult cancer survivors and their associated health and QOL outcomes including physical functioning (strength, activities of daily living, fatigue, etc.) and body composition (changes in weight, percent body fat, etc.).

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Methods:

Study selection criteria

PubMed and Web of Science databases were searched due to their emphasis on scientific literature to ensure all relevant literature was included in the review. Studies that met all of the following PICOS (Table 1) criteria were included in the review: (1) studies must include human subjects with a pediatric, adolescent, or young adult cancer diagnosis (ages 0 to 39); (2) full-text articles must have appeared in peer-reviewed journals; (3) papers must have been published in English; (4) studies must be quantitative in nature; (5) studies must have reported at least one physical functioning, body composition, or QOL outcome; (6) studies must include a physical activity component; and (7) studies must have been a randomized clinical trial (RCT) design. Studies were identified through structured searches of January 2018 – June 29, 2021. Studies were excluded from review if meeting one or more of the following criteria: (1) studies not written in English; (2) studies reporting malignancies other than cancer; (3) if cancer survivors were equal to or greater than 40 years of age at diagnosis; (4) grey literature such as reviews, letters, editorials, dissertations, and conference abstracts; and (5) study designs other than a RCT.

A time restriction of four years was applied to the publication date of papers as a rigorous scoping review examined physical activity interventions up to and including those published in 2017 in cancer survivors diagnosed between 15 to 39 years of age [19]. Since 2017, there has been an increase in high-quality research adopting physical activity RCT study designs in pediatric, adolescent, and young adult cancer survivors. The purpose of this study was to expand on the current state of the literature by focusing on RCTs, the gold standard for effectiveness research, and expanding the population to include pediatric cancer survivors.

Search Strategy

To ensure transparency and complete reporting, the systematic review protocol described in the Preferred Reporting Items for Systematic Reviews and Meta-Analysis statement was adopted to guide the review process [24, 25]. A comprehensive keyword search was performed in PubMed and Web of Science that included objective measures of physical functioning, physical activity, and objective assessment measures. The search algorithm included all possible combinations of keywords (“child” or “adolescent” or “pediatric” or “kid” or “teen” or “young adult” or “surviv” or “patient”) AND (“cancer” or “oncology” or “neoplasms” or “tumor”) AND (“physical” or “physical activity” or “activity” or “activ” or “exercise” or “exercis” or “run” or “walk” or “jog” or “physical fitness” or “swimming” or “fitness class” or “fitness program” or “fitness” or “tai chi” or “gardening” or “physical education” or “physical training” or “dancing” or “sport” or “pilates” or “yoga” or “cycle” or “cycling” or “bike” or “movement” or “pedometer” or “tracker” or “physical intervention” or “Fitbit” or “accelerometer” or “steps” or “distance” or “actigraphy”).

To determine the relevance of all potentially pertinent studies, all titles and abstracts of the articles identified through keyword search were screened against the study inclusion and exclusion criteria. Potentially relevant articles were retrieved for evaluation of the full text.

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A cited reference search was performed based on the articles meeting the study selection criteria identified from the keyword search. Articles identified were further screened and evaluated using the same study selection criteria. Two authors of this review (SC and AB) jointly determined the inclusion/exclusion of all articles retrieved in full text, and discrepancies were resolved through discussion.

Data extraction

Data extraction was used to collect the following methodological and outcome variables from each included study: primary investigator(s), country of publication, publication year, purpose/aim of study, sample demographics (population and age), cancer type, main results/findings, and physical functioning, body composition, and QOL outcome(s) (Tables 2, 3, 4).

Data analysis

Two authors, S. Crowder and A. Buro, independently read and recorded the details of the selected studies. Each selected article was analyzed for key findings. Hypothesized *a priori* study outcomes are reported accordingly. Traditionally, two studies are sufficient to perform a meta-analysis, provided that those two studies can be meaningfully pooled and provided their results are sufficiently 'similar'. However, in our review, no two studies provided corresponding quantitative results; therefore, a meta-analysis was not possible [26].

Study quality assessment

Study quality was adapted from a previous systematic review [27] and was assessed by the following criteria [28]: (1) Was the research question/hypothesis clearly stated? (2) Were the inclusion and exclusion criteria clearly stated? (3) Was a clear aim for the study stated? (4) Were the methods of the study clearly described? (5) Were the main findings (results) of the study clearly described? (6) Were study limitations discussed? (7) Were physical activity recommendations in agreement with guidelines (e.g., American Cancer Society (ACS) or the American Institute of Cancer Research (AICR))? (8) Was a sample size justification via power analysis provided? (9) Are data analyses discussed? Two authors of the review independently scored each study based on these criteria, with dis-agreement resolved through discussion. Scores for each criterion range from 0 to 2, depending on whether the criterion was unmentioned or unmet (0), partially met (1), or completely met (2). The possible total study score ranged between 0 and 18. A higher total score represents higher quality. Study quality score assisted in the measure of the strength of study evidence; therefore, the purpose was not to exclude studies but rather to help the researchers gain a deeper understanding of each one. This was an overall representation of the entire systematic review quality for all articles used as compared to assessing each individual article.

Results

Study selection

PubMed and Web of Science database searches yielded 4,770 studies. Following the removal of duplicates and title and abstract screening, 83 full-text articles were assessed, and 9 studies met the inclusion criteria. In the PRISMA flowchart (Figure 1), the stages of the review process, including study identification, inclusion, and exclusion, are shown. Articles

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excluded after full text review with reason included: (1) not a RCT (n=8); (2) outside of the 0-39 age group (n=65); and (3) no cancer diagnosis (n=1). The selected studies were published between 2018 and 2021 and their study outcomes are presented in Table 2 and study characteristics in Table 3. An overview of the included studies is summarized in Table 4.

Overview characteristics of selected studies

Table 4 reports basic characteristics of the nine articles included in the review. Studies were conducted across eight countries, including the United States [29, 30], China [31], Finland [32], Spain [33], Netherlands [34], Australia [35], France [36], and Taiwan [12]. The majority of studies had a sample size between 26 to 50 survivors [29, 32, 35]. Four studies justified their physical activity recommendations using recommended guidelines including the National Strength and Conditioning Association guidelines [29], the American College of Sports Medicine and/or Exercise and Sports Science Australia guidelines [33, 35], and the Center for Disease Control and Prevention Physical Activity Guidelines [30]. Three studies included a theoretical framework including Kolb's experiential learning theory [31], social cognitive theory [29], and self-efficacy theory [12]. Seven studies included childhood and adolescent cancer survivors (0-18 years) [12, 30-34, 36] and two studies included young adult cancer survivors (18-39 years) [29, 35]. The physical activity interventions ranged from one week [12], 8 weeks [32], 10 weeks [35], 12 weeks [29, 34], 16 weeks [33], 24 weeks [30], and across six months [31, 36]. Intervention modalities varied across all nine studies. In a feasibility study by Devine et al., eight in-person group exercise sessions with mobile app and Fitbit followed by 4 weeks of mobile app and Fitbit only was evaluated [29]. Hamari et al. evaluated the effects of playing the Nintendo Wii™ Fit for 30 min/day for 8 weeks on physical activity, motor performance, and fatigue [32]. In the study by Wu et al., the acceptability and effectiveness of a tailored education on health behaviors was evaluated, including six 45-60 minute individual education and follow-up telephone counseling for one week [12]. The study by Manchola- Gonzalez et al., evaluated the effects of a home-exercise program on physical fitness indicators and physical functioning at baseline and over a 16-week intervention [33]. Atkinson et al., assessed cardiorespiratory fitness on an exercise group receiving a 10-week program comprising progressive aerobic and resistance exercise [35]. An efficacy study by Howell et al. assessed a web-delivered physical activity intervention to increase moderate-to-vigorous physical activity and improve fitness over 24 weeks [30]. Li et al. assessed the effectiveness of an adventure-based training program in promoting physical activity by undergoing a 4-day program, with one training day at 2 weeks, 2, 4, and 6 months after randomization [31]. Braam et al. evaluated a 12-week combined physical exercise and psychosocial intervention [34]. Finally, Saultier et al. evaluated the safety and efficacy of a 6-month physical activity program, including 30 sessions of 30-to-90 minute muscle building and balance training and 15 multi-activity sessions of 90 to 240 minutes including dance, basketball, yoga, etc. [36]. Sessions for the 6-month physical activity program were performed during 20 days of physical preparation (department gym, patient's room, or outdoors) and three stays, including two weekend stays and one long stay of five consecutive days [36].

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Outcomes of interest

Physical activity outcomes of interest were selected in the initial review process *a priori* and included physical functioning (e.g., strength, fatigue, activities of daily living), measures of body composition (e.g., changes in weight, percent body fat) and QOL (e.g., total and domain specific QOL) and are represented in the *a priori* hypothesized theoretical model illustrated in Figure 2. Physical activity is located in the center of the model and hypothesized associations are represented with bidirectional arrows.

Physical activity

Physical activity was analyzed using self-reported questionnaires in six studies including The Chinese University of Hong Kong Physical Activity Rating for Children and Youth (CUHK-PARCY) [31], the metabolic equivalent questionnaire (MET) [32], activity diaries [32], the Godin-Shephard Leisure Time Physical Activity Questionnaire (GSLTPAQ) [35], the Modified International Physical Activity Questionnaire-Short Form (IPAQ-SF) [29], the Physical Activity Questionnaire for Adolescents (PAQ-A) [33], and the Healthy Behavior Self-Efficacy (HBSE) and Health Promotion Lifestyle (HPL) questionnaires [12]. Objective measures of physical activity were collected in five studies including accelerometers of various brands i.e. Fitbits [29, 32], ActiGraphs [29, 30], Actical accelerometers [34], and Polar heart rate monitors [36]. Across all studies reviewed, two studies reported improvements in physical activity [31, 36]. In the study by Li et al., the experimental group showed statistically significantly higher levels of physical activity than the control group ($p<0.001$) [31]. In the study by Saultier et al., physical activity in the intervention group significantly improved exercise capacity and physical parameters ($p=0.0004$) [36]. Five studies reported partial improvements [29, 30, 33-35], and two studies reported no improvements [12, 32].

Physical functioning

Measures of physical functioning included fatigue, strength, cardiovascular fitness, flexibility, and activities of daily living. Fatigue was assessed subjectively using the Chinese version of the Fatigue Scale-Child in one study [31], the Pediatric Quality of Life Fatigue Scale (PedsQL-F) in three studies [29, 32, 34], and the Functional Assessment of Chronic Illness Therapy-Fatigue (FACT-F) in one study [35].

Strength was assessed objectively using a variety of methodologies including handgrip strength [30, 33-35], 10-repetition maximum tests for lower body (leg press machine) and upper body (barbell) following the National Strength and Conditioning Association guidelines [29], lower-body hip, knee and ankle dorsiflexion score [34], back and leg strength using a dynamometer [35], maximal push-up and sit-up tests [30, 35], and a Eurofit battery incorporating flexibility, balance, upper limb strength, lower limb strength, and trunk/abdominal endurance [36].

Cardiorespiratory fitness was measured in four studies via aerobic biomarkers (e.g., peak oxygen uptake (VO_2 max)) [29, 33-35]. Flexibility was measured by the sit-and-reach test [33, 35] and the back scratch test [35]. Activities of daily living included the timed up and

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go test (TUG) and the timed up and down stairs test (TUD) [33], the six minute walk test [36], and the movement assessment battery for children [32].

Significant improvements in physical functioning were noted in five studies [29, 31, 33, 35, 36]. In repeated measures ANOVA, there was a significant group x time effect for lower body muscle strength, with the intervention group demonstrating a greater increase post-intervention (30.8% vs. 11.7% in wait list controls p=0.02) [29]. However, the findings were not significant at 6 months and there were no significant improvements in upper body strength or cardiorespiratory fitness postintervention or 6 months [29]. In a study by Manchola-Gonaelez, in adjusted mixed linear models, there was a significant group x time interaction for VO_2 peak (p=0.03). In the same study, changes in mean values were observed in minute ventilation (p=0.03), peak oxygen uptake (p=0.04), maximal load (p=0.02), timed up and down stairs test (p=0.01), and timed up and go test (p=0.001) however the group x time interaction was not significant [33]. Similarly, in a study by Atkinson et al., the exercise group demonstrated significant improvement in VO_2 peak at 10 weeks compared with controls (p=0.0002), but by six months, the difference was no longer significant (p=0.21) [35]. In a study by Saultier et al., the six-minute walk test improved in the intervention group vs. the control group (p<0.001) [36]. In a study by Li et al., the experimental group reported statistically significantly lower levels of cancer-related fatigue than the control group (p<0.001) [31]. Nonsignificant improvements or no improvements were noted in three studies [30, 32, 34]. Physical functioning was not a reported outcome in one study [12].

Body composition

Body composition was a reported outcome in two studies. In one study, it was assessed using a Dual-energy-X-ray Absorptiometry (DXA)-scanner to assess fat mass percentage and lumbar spine [34]. At 12 months follow-up, significant improvements in lower body muscle strength were noted in the intervention group when compared to the control group (β = 56.5 Newton; 95% C = 8.5; 104.5) [34]. In a study by Devine et al., body composition was assessed using changes in BMI or waist circumference, however there were no statistically significant changes postintervention or at 6 months follow-up [29].

Quality of life

Quality of life was assessed using the Pediatric Quality of Life Inventory (PedsQL) in five studies [29-31, 34, 35], the parent-reported version of the “Vécu et Santé Perçue de l’Adolescent et de l’enfant” questionnaire [36] and the Healthy Behavior Self-Efficacy (HBSE) and Health Promotion Lifestyle (HPL) questionnaires [12]. Significant improvements in QOL were noted in two studies [31, 36]. In the study by Li et al., the experimental group showed statistically significantly higher levels of self-efficacy (p<0.001) and better QOL (p <0.01) [31]. In the study by Saultier et al., changes in global self-esteem significantly improved in the intervention group (p=0.04) and parent reported QOL change was better in the intervention group (p=0.04) from baseline to 6 and 12 months [36]. Nonsignificant improvements or no improvements were noted in five studies [29, 30, 32, 34, 35]. QOL was not a reported outcome in two studies [32, 33].

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Study quality assessment

Table 5 reports the results of the study quality assessment and the methodology used to analyze collected data. Studies included in the review scored 13 out of 18 on average and ranged between 10 and 16. All 9 of the analyzed research studies clearly described the research methodologies. Four studies used physical activity guidelines to guide their intervention. Six of the selected studies clearly considered the studies' limitations and weaknesses.

Discussion

Extensive literature highlights the benefits of physical activity in managing symptoms and promoting health in adult cancer survivors [37-39], however, pediatric, adolescent and young adult cancer survivors remain disproportionately underrepresented in the literature. Considering pediatric, adolescent and young adult cancer survivors have needs far different than adult cancer survivors, the number of studies enrolling these patients in recent years has increased as an attempt to address this issue. Adolescent and young adult cancer survivors report lower physical activity during and after treatment, thus there is a critical need to design behavior change interventions to improve health-related outcomes [40]. Our research team conducted a systematic review of the literature to summarize the impact of physical activity interventions in adolescent and young adult cancer survivors and identify knowledge gaps that warrant further investigation.

This systematic review included 9 studies in pediatric, adolescent and young adult cancer survivors and extended upon a recent scoping review published in 2017 [19]. Although the 2017 scoping review used rigorous and transparent methods, scoping reviews are conducted for different purposes than a systematic review, such as discussing concepts rather than answering a specific research question and provide a broad scope of a body of literature [41]. For this review, we systematically extended upon the original scoping review, and similar to that review concluded that physical activity is not well researched among pediatric, adolescent, and young adult cancer survivors. This review highlighted the need for objective measures of strength and body composition in future physical activity studies. For instance, only two studies assessed body composition, therefore future research to assess objective measures of body composition are warranted. Additionally, physical activity was commonly assessed using self-reported measures. While self-reported measures are commonly used and validated in the population, they are susceptible to recall and social desirability bias. Bias in self-reported questionnaires may influence the relationship between physical activity and health-related outcomes [19]. Therefore direct, objective measures of physical activity are warranted for future research studies in this population. Finally, the studies lacked exploration of outcomes known to be important to adolescent and young adult cancer survivors (e.g., hindering fatigue and promoting cognitive function/mindfulness) [16, 42]. Mindfulness based programs may eliminate physical and psychological barriers to activity while increasing motivation for exercise and RCT adherence [12, 42].

An unexpected finding of the review was that only two publications included adolescent and young adult cancer survivors, warranting a need for further research in this population. The initial goal of the review was to assess pediatric as compared to adolescent and young adult

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cancer survivors. However, limited research was available in young adult cancer survivors. Upon further exploration, young adults with cancer likely have far different preferences and needs than childhood and adolescent cancer survivors. Therefore, physical activity interventions for young adult cancer survivors, specifically for those 20 years of age and older, are needed. It is possible combining adolescent and young adult cancer survivors together in one study could mask an effect that is stronger for one subgroup (e.g., activity during school hours vs. activity during a workday) or could suggest an effect applies to only one subgroup (e.g., adolescence vs. young adulthood) [19].

Upon completion of this systematic review, the types of physical activity programs preferable to survivors, timing of programs, and preferences for programming within the cancer trajectory remain undetermined. Further multidisciplinary work collaborating with adolescent and young adult cancer survivors to obtain their perspectives on unmet needs and health concerns are encouraged when designing physical activity programs to address knowledge gaps to enhance optimal programming.

Limitations of the review should be highlighted. First, studies were included if they were published in English, which could limit the generalizability of findings globally. Second, although we aimed to create a comprehensive search covering literature across multiple databases, our review may not encompass all available research. Third, the heterogeneity of the studies (e.g., variations in intervention length, outcomes, age, cancer type, time since treatment) prevented the ability to conduct a meta-analysis. Therefore, we do not have the ability to draw firm conclusions.

Conclusion

In conclusion, while efforts to increase health-related adolescent and young adult research exist, there is a lack of consistent findings across studies. Across all studies reviewed, two reported improvements in physical activity, five reported partial improvements, and two reported no improvement. Only a handful of studies are available that explore the effects of physical activity on health-related outcomes including physical functioning, QOL, and body composition, highlighting a considerable knowledge gap. Furthermore, only two physical activity interventions incorporated young adults with cancer, who likely have different preferences and needs than childhood and adolescent cancer survivors. Thus, physical activity interventions for young adult cancer survivors, specifically 20 years of age and older, should be explored. Future research should focus on personalized physical activity components encouraging behavior change techniques with objective measures of physical functioning and body composition.

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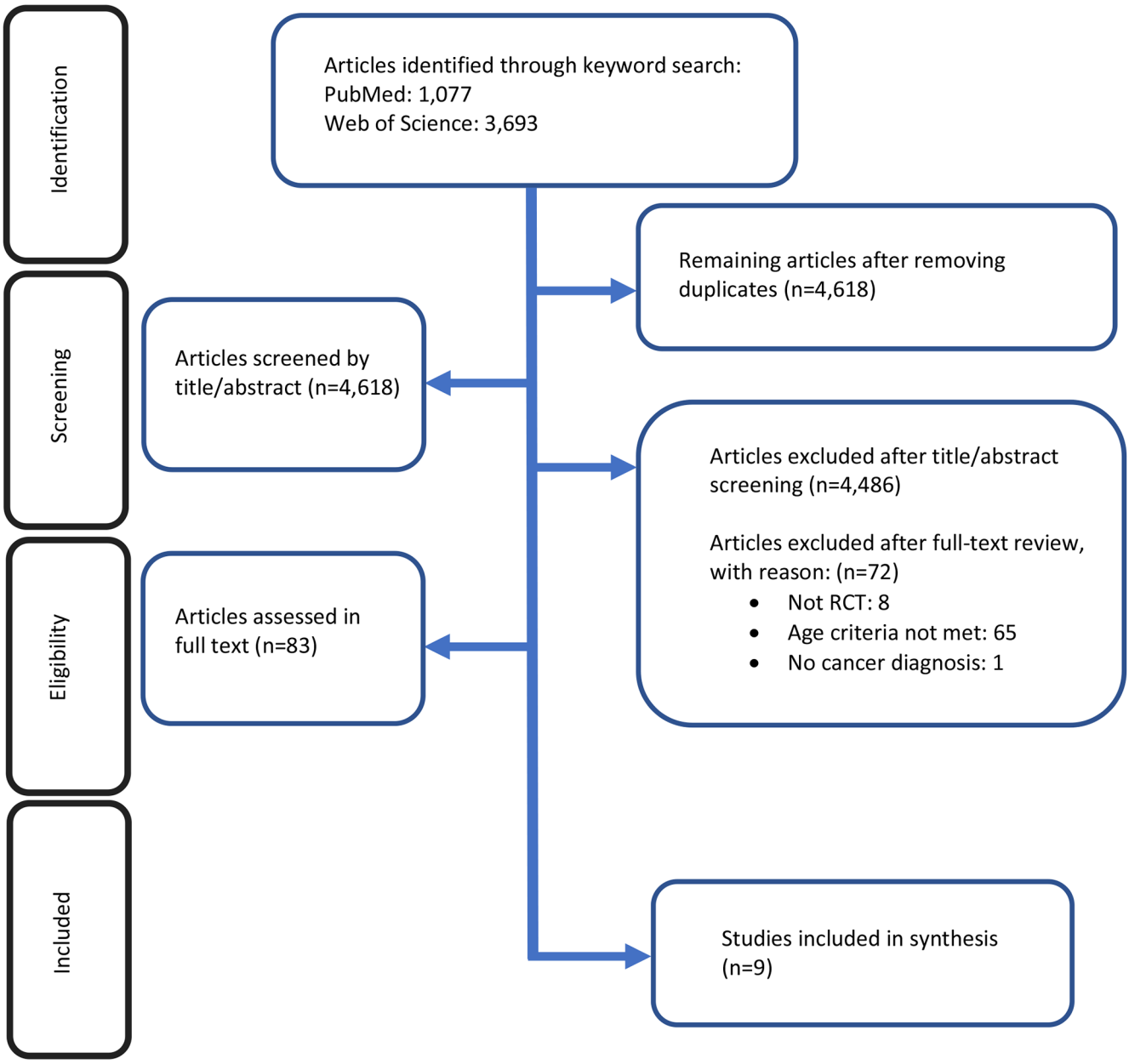

Figure 1: PRISMA FLOWCHART

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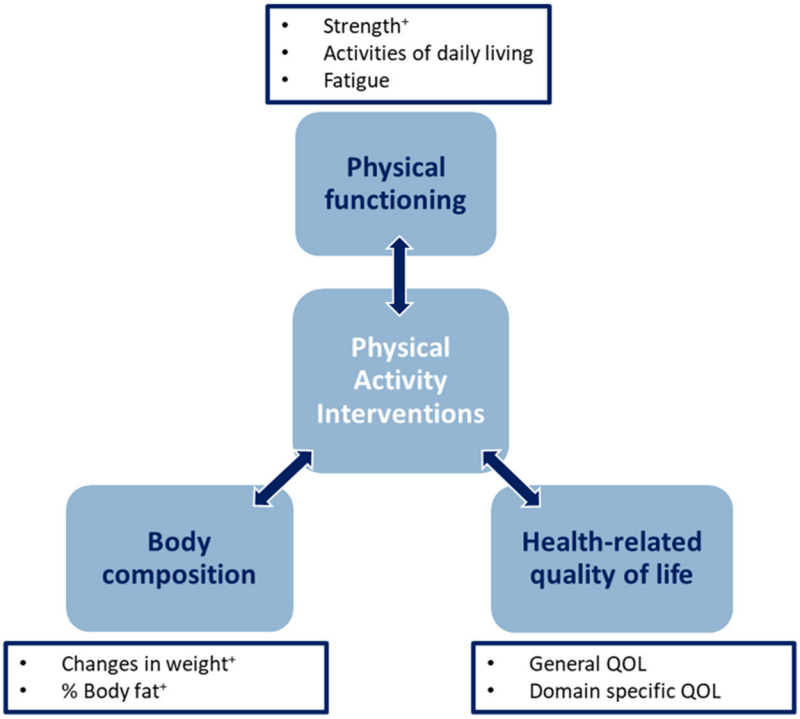

Figure 2: Theoretical model- *a priori* hypothesized outcomes of physical activity interventions

Theoretical model—a priori hypothesized outcomes of physical activity interventions.

+Measures in this category were not comprehensively addressed in the reviewed studies. These illustrate a gap in current research.

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Table 1:

PICOS characteristics

Question: **Physical activity interventions and associated outcomes in pediatric/adolescent young adult cancer survivors**		
Acronym	**Definition**	**Description**
P	Patient or problem	Pediatric and adolescent young adult cancer survivors
I	Intervention	Physical activity intervention
C	Comparison	Not applicable
O	Outcomes	Any
S	Study design	Randomized Clinical Trial

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Table 2:

Study Outcomes

Descriptive Information	Physical Functioning (ex. Strength, activities of daily living, fatigue)	Body Composition (ex. % body fat)	Quality of Life (ex. Self-reported questionnaires)	Physical Activity measurement and length
Author/[Reference]: Li [31] Publication Year: 2018 Participant Age: 9-16 Cancer Type: Leukemia, lymphoma, brain, bone, neuroblastoma Post-Treatment: 6 to > 60 months	Subjective: Chinese version of the FS-C to assess fatigue	NA	Subjective: Quality of life was assessed using the Chinese version of the PEDSQL	Subjective: CUHK-PARCY to assess physical activity Data collection occurred at baseline, 6 and 12 months Total of 4-day adventure-based training program: One day 2 weeks after randomization and one day at 2, 4, and 6 months
Author/[Reference]: Devine [29] Publication Year: 2020 Participant Age: 13-25 Cancer Type: Blood, brain, solid tumor Post-Treatment: Average 9 years	Objective: Lower and upper body muscular strength was measured via 10-repetition maximum tests following the National Strength and Conditioning Association guidelines (leg press machine for lower body and barbell bench press for upper body) Cardiorespiratory fitness was measured via submaximal treadmill test Subjective: PedsQL fatigue scale was used to measure fatigue	Objective: Changes in BMI and waist circumference measures	Subjective: PedsQL	Objective: Fitbit Charge or Alta and the ActiGraph wGT3X-BT accelerometer. Subjective: Self-reported minutes of moderate-vigorous physical activity using the modified International Physical Activity Questionnaire-Short Form. Sedentary behavior was measured using two items from the PACE adolescent psychological and sedentary behavior survey. 12-week intervention including 8 weeks of in-person group sessions followed by 4 weeks of mobile app and fitness tracker use alone
Descriptive Information	**Physical Functioning (ex. Strength, activities of daily living, fatigue)**	**Body Composition (ex. % body fat)**	**Quality of Life (ex. Self-reported questionnaires)**	**Physical Activity measurement and length**
Author/[Reference]: Manchola-Gonzalez [33] Publication Year: 2019 Participant Age: 7-17 Cancer Type: ALL Post-Treatment: Minimum 1 year	Objective: Cardiopulmonary fitness (VO_{2peak}) using a metabolic cart. Pulse oxygen, output of carbon dioxide, minute ventilation, respiratory exchange rate, and anaerobic threshold Objective: Strength- maximal strength was assessed by handgrip strength (Jamar) Objective: Flexibility was measured by the sit-and-reach test using procedure outlined in the American College of Sports Medicine manual Objective: Activities of daily living-timed up and go test (TUG). Timed Up and Down Stairs test (TUDS)	NA	NA	Subjective: Physical Activity Questionnaire for Adolescents (PAQ-A) 16-week intervention
Author/[Reference]: Braam [34] Publication Year: 2018 Participant Age: 8-18 Cancer Type: ALL, AML, HL, non-HL, CML, Burkitt, CNS/brain, solid tumor Post-Treatment: During treatment or within 12-months post-treatment	Objective: Cardiorespiratory fitness (VO_{2peak}) cardiopulmonary exercise test using the Godfrey protocol Objective: Strength-using a hand-held dynamometer and lower-body hip, knee and ankle dorsiflexion score Subjective: PedsQL fatigue scale was used to measure fatigue	Objective: Percent fat mass and lumbar spine as measured by DXA	Self-reported: Psychosocial function and QOL- QOL measures using the Dutch self-report version of the PedsQL, Children's depressive Inventory, athletic competence and global self-worth of the "self-perception	Objective: PA measured using Actical accelerometer (objective) 12-week intervention

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Descriptive Information	Physical Functioning (ex. Strength, activities of daily living, fatigue)	Body Composition (ex. % body fat)	Quality of Life (ex. Self-reported questionnaires)	Physical Activity measurement and length
			profile", behavioral problems reported on the Youth self-report	
Descriptive Information	**Physical Functioning (ex. Strength, activities of daily living, fatigue)**	**Body Composition (ex. % body fat)**	**Quality of Life (ex. Self-reported questionnaires)**	**Physical Activity measurement and length**
Author/[Reference]: Saultier [36] Publication Year: 2021 Participant Age: 5-18 Cancer Type: Leukemia, brain, lymphoma, bone, other solid tumor Post-Diagnosis: Average 10.7 months	Objective: Exercise capacity using six minute walk test Objective: Eurofit test battery-flexibility (sit and reach), balance (flamingo balance test), upper limb strength (medicine ball launch), lower limb strength Myotest and chair test), trunk muscle endurance (bridge trunk muscle endurance test), and abdominal muscle endurance (sit-up score)	NA	Subjective: Physical self-inventory-very short form Subjective: Vécu et Santé Perçue de l'Adolescent et de l'enfant questionnaire	Objective: Heart rate monitor (V800, Polar) 6 months
Author/[Reference]: Wu [12] Publication Year: 2019 Participant Age: 8-20 Cancer Type: ALL, AML, Lymphoma, Brain, LCH, solid tumor Post-Treatment: within ± 2 months completing treatment	NA	NA	Subjective: HBSE and HPL questionnaire	Subjective: HBSE and HPL questionnaire 1-week intervention
Author/[Reference]: Hamari [32] Publication Year: 2019 Participant Age: 3-16 Cancer Type: ALL, Burkitt, HL, non-HL, other neoplasm Post-Diagnosis: Mean 13.3 days from initial diagnosis	Objective: Movement-ABC2 for motor performance Subjective: PedsQL fatigue scale was used to measure fatigue	NA	NA	Objective: Assessed with accelerometers (Fitbit Ultra) Subjective: Activity diaries (min/day), metabolic equivalent (MET) questionnaire to assess leisure-time PA in hours/week, and questionnaires 8-week intervention
Author/[Reference]: Howell [30] Publication Year: 2018 Participant Age: ≥11 to < 15 Cancer Type: ALL, AML, CNS, Sacroma, HL, non-HL, other neoplasm Survival Time: Mean 9.3 years	Objective: Strength-maximal strength was assessed by handgrip strength (Jamar) Objective: Proximal muscle strength-full sit ups and either a knee of full push-up for 30 seconds	NA	Subjective: PedsQL	Objective: Baseline, 12 and 24 weeks using triaxial accelerometer (Wgt3x-bt; ActiGraph) 24-week intervention

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Table 3:

Study Characteristics

Descriptive Information	Aim	Intervention	Outcome
Author/[Reference]: Li [31] Publication Year: 2018 Participant Age: 9-16 Cancer Type: Leukemia, lymphoma, brain, bone, neuroblastoma Post-Treatment: 6 to > 60 months	Examine the effectiveness of an adventure-based training program in promoting physical activity, reducing fatigue, and enhancing self-efficacy and quality of life among Hong Kong Chinese childhood cancer survivors.	The experimental group underwent a 4-day adventure-based training program. The control group received a placebo intervention. The primary outcome was fatigue at 12 months. Secondary outcomes were physical activity levels, self-efficacy and quality of life at 12 months. Data collection was conducted at baseline, 6 and 12 months after the intervention began.	There were 222 eligible childhood cancer survivors assigned to either an experimental (n = 117) or a control group (n = 105). The experimental group showed statistically significantly lower levels of cancer-related fatigue ($P < 0.001$), higher levels of self-efficacy ($P < 0.001$) and physical activity ($P < 0.001$), and better quality of life ($P < 0.01$) than the control group at 12 months.
Author/[Reference]: Devine [29] Publication Year: 2020 Participant Age: 13-25 Cancer Type: Blood, brain, solid tumor Post-Treatment: Average 9 years	To evaluate the feasibility of a technology-enhanced group-based fitness intervention for AYA survivors of childhood cancer.	AYA survivors ages 13-25 years were randomized to the intervention (eight in-person group sessions with mobile app and FitBit followed by 4 weeks of app and FitBit only) or waitlist control. Assessments were at 0, 2, 3, 6, and 9 months. Feasibility was evaluated by enrollment, retention, attendance, app engagement, and satisfaction. Secondary outcomes included physical activity, muscular strength/endurance, cardiorespiratory fitness, health-related quality of life, and fatigue.	Forty-nine survivors (M_{age} = 18.5 years, 49% female) completed baseline assessments and were randomized (25 intervention, 24 waitlist). Thirty-seven (76%) completed the postintervention assessment and 32 (65%) completed the final assessment. On average, participants attended 5.7 of eight sessions (range 1-8). Overall intervention satisfaction was high (M = 4.3, SD = 0.58 on 1-5 scale). Satisfaction with the companion app was moderately high (M = 3.4, SD = 0.97). The intervention group demonstrated significantly greater improvement in lower body muscle strength compared to the waitlist post-intervention, and small but not statistically significant changes in other secondary measures.
Descriptive Information	**Aim**	**Intervention**	**Outcome**
Author/[Reference]: Manchola-Gonzalez [33] Publication Year: 2019 Participant Age: 7-17 Cancer Type: ALL Post-Treatment: Minimum 1 year	The aim of this study was to evaluate the effects of a home-exercise program on physical fitness indicators and physical functioning after completion of chemotherapy in children and adolescents diagnosed with acute lymphoblastic leukemia (ALL).	Twenty-four survivors of ALL were assigned to usual care (control group, n = 12, 11.0 ± 3.7 years) or to a home-exercise program (intervention group, n = 12, 11.8 ± 4.3 years). Peak oxygen uptake (VO_2peak ml/kg/min), minute ventilation (VE L/min), output of carbon dioxide (VCO_2 L/min), respiratory exchange ratio (RER), peak heart rate (beats/min), maximal load (W), VO_2 at anaerobic threshold (VO_2 at AT, ml/kg/min), pulse oxygen (PO_2 ml/beat), heart rate at anaerobic threshold (beats/min), handgrip test (pounds), flexibility (cm), Timed Up & Go test TUG (s), and Timed Up and Down Stairs test (TUDS s) were measured at baseline and over 16 weeks of intervention.	Adjusted mixed linear models revealed a significant group-time interaction + 6.7 (95% CI = 0.6-12.8 ml/kg/min; η^2 partial = 0.046, P = 0.035) for VO_2peak. Similarly, changes in mean values were observed after the home-exercise program compared with baseline for VE (L/min) - 8.8 (3.0) (P = 0.035), VCO_2 - 0.2 (0.08), (P = 0.041), maximal load (W) - 35.5 (12.8) (P = 0.024), TUDS (s) 0.8 (2.6) (P = 0.010), and TUG (s) 0.6 (0.1) (P = 0.001); however, the group-time interaction was not significant.
Author/[Reference]: Braam [34] Publication Year: 2018 Participant Age: 8-18 Cancer Type: ALL, AML, HL, non-HL, CML, Burkitt, CNS/brain, solid tumor Post-Treatment: During treatment or within 12-months post-treatment	Evaluate the effect of a combined physical exercise and psychosocial intervention on cardiorespiratory fitness, muscle strength, body composition, psychosocial function and health-related quality of life (HrQoL).	The 12-week intervention consisted of 24 individual physical exercise sessions supervised by a physiotherapist, and 6 psychosocial training sessions for children and 2 for parents. Physical fitness and psychosocial function were assessed at baseline, directly post-intervention and at 12 months' post-baseline. Generalized estimating equations were used to simultaneously assess intervention effects at short and long-term.	No significant differences in the effects of the intervention were found on physical fitness and psychosocial function at short-term in intervention vs. control. At 12-months follow-up, significantly larger improvements in lower body muscle strength (β = 56.5 Newton; 95% CI: 8.5; 104.5) were found in the intervention group when compared to the control group. Within-group changes showed significant improvements over time in HrQoL and bone density in both groups. Intervention effects on HrQoL were not significantly mediated by physical fitness and psychological function.
Descriptive Information	**Aim**	**Intervention**	**Outcome**

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Descriptive Information	Aim	Intervention	Outcome
Author/[Reference]: Saultier [36] Publication Year: 2021 Participant Age: 5-18 Cancer Type: Leukemia, brain, lymphoma, bone, other solid tumor Post-Diagnosis: Average 10.7 months	Evaluate the safety and efficacy of a physical activity program in children and adolescents with cancer.	Children and adolescents with cancer were randomly assigned in a 1:1 ratio to the six-month physical activity program (intervention group) or to the control group. The first evaluation was performed at the end of the PAP (T0 + 6 mo). At T0 + 6 mo, both groups received the six-month PAP with a second evaluation at T0 + 12 mo. The primary outcome was the evolution of exercise capacity measured using the six-minute walk test (6 MWT) at T0 + 6 mo. Secondary outcomes included physical activity program safety and changes in other physical functions, self-esteem, and quality-of-life parameters.	At T0 + 6 mo, the evolution of the 6 MWT distance (±SEM) was improved in the intervention group vs. the control group (86 ± 12 m vs. 32 ± 6 m, $p < 0.001$). Several other physical parameters were significantly improved in the intervention group. Global self-esteem and parent-reported quality-of-life were significantly increased in the intervention group. Analysis at T0 + 12 mo showed persistence of the benefits in the intervention group on exercise capacity evolution (115 ± 18 m vs. 49 ± 11 m, $p = 0.004$) and on most physical and QoL parameters.
Author/[Reference]: Wu [12] Publication Year: 2019 Participant Age: 8-20 Cancer Type: ALL, AML, Lymphoma, Brain, LCH, solid tumor Post-Treatment: within ± 2 months completing treatment	Evaluate the acceptability and effectiveness of a tailored education on healthy behavior self-efficacy (HBSE) and health promotion lifestyle (HPL) for childhood cancer survivors.	A two-group, randomized study with repeated measures was conducted in Taiwan. Participants were randomly assigned to receive six 45-60 min individual education and follow-up telephone counselling sessions (n = 34) or standard of care only (n = 35). Each participant was assessed with HBSE and HPL questionnaires and was evaluated at three time points (at baseline, and then 1 and 4 months after intervention).	HBSE and HPL scores increased across the three time points in the experimental group (all $p < 0.05$), except for the HBSE exercise subscale ($p = 0.85$). HBSE scores were significantly higher for the experimental group than for the control group after 4 months of intervention ($F = 5.32$, $p = 0.02$, $\eta^2 = 0.25$). No significant improvements in HBSE were observed over time in the control group.
Descriptive Information	**Aim**	**Intervention**	**Outcome**
Author/[Reference]: Hamari [32] Publication Year: 2019 Participant Age: 3-16 Cancer Type: ALL, Burkitt, HL, non-HL, other neoplasm Post-Diagnosis: Mean 13.3 days from initial diagnosis	Evaluate the effect of active video games in promoting physical activity and motor performance, and reducing fatigue in children with cancer.	The intervention included playing Nintendo Wii™Fit (Nintendo Co., Ltd., Kyoto, Japan) for 30 min/day for 8 weeks. Physical activity was estimated with accelerometers, physical activity diaries and questionnaires. Movement-ABC2 and PedsQL™ were used to examine motor performance and fatigue.	Participants were randomly assigned to the intervention and control groups. The median [min-max] accelerometer counts/h (500 [131-1130] vs 385 [116-1012], p = 0.63) and physical activity min/day (34 [0-150] vs 23 [0-260], p = 0.95) did not differ between the groups. Change between the pre-test and post-test regarding motor performance and fatigue was similar in both groups (motor performance p = 0.77; fatigue p = 1.00). Overall, the physical activity levels were low and one fourth of the children had or were at risk of having movement difficulties.
Author/[Reference]: Howell [30] Publication Year: 2018 Participant Age: ≥11 to < 15 Cancer Type: ALL, AML, CNS, Sacroma, HL, non-HL, other neoplasm Survival Time: Mean 9.3 years	Evaluate the efficacy of a web-delivered physical activity intervention among adolescent survivors to increase moderate to vigorous physical activity (MVPA) and improve fitness and neurocognitive and health-related quality of life (HRQoL) over 24 weeks.	Participants were randomized to either a physical activity intervention delivered over the internet or a control group. The intervention group received educational materials, an activity monitor, and access to an interactive website designed to motivate increased physical activity via rewards; the control group received an activity monitor and educational materials. Physical activity, fitness, and neurocognitive and HRQoL outcomes were assessed at baseline and at 24 weeks. Mean changes were compared between groups using paired t-tests.	While survivors in the intervention group increased, and those in the control group decreased (4.7 ± 119.9 vs. −24.3 ± 89.7 min) weekly MVPA, this difference was not significant (P = 0.30). However, hand grip strength, number of sit-ups and pushups, neurocognitive function, and HRQoL outcomes improved in the intervention, but not in the control group.

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Table 4:

Overview of Included Studies

All included studies (N = 9)		
Total number of study participants (N = 349 experimental group; N = 310 control group)		
Gender (N = 293 female; N = 347 male)*		
Age of participants in years; Mean (range): 13.3 (3 - 25)		
Study Characteristics	**Categories**	**Number of Studies**
Country	USA	2
	China	1
	Finland	1
	Spain	1
	Netherlands	1
	Australia	1
	France	1
	Taiwan	1
Year of publication	2018	3
	2019	2
	2020	2
	2021	2
Mean age of cancer survivor per study	Childhood and adolescence (0-18 years)	7
	Young adult (18 – 39 years)	2
Total sample size	≤25	1
	26-50	3
	51-75	2
	76-100	2
	≥101	1
Physical activity guidelines	National Strength and Conditioning Guidelines	1
	American College of Sports Medicine	2
	Center for Disease Control and Prevention Physical Activity Guidelines	1
	No mention of guidelines	5
Length of intervention	1 week	1
	8 weeks	1
	10 weeks	1
	12 weeks	2
	16 weeks	1
	24 weeks	1
	Across 6 months	2
Theoretical framework	Kolb's experiential learning theory	1
	Social cognitive theory	1

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All included studies (N = 9)		
Total number of study participants (N = 349 experimental group; N = 310 control group)		
Gender (N = 293 female; N = 347 male)*		
Age of participants in years; Mean (range): 13.3 (3 - 25)		
Study Characteristics	**Categories**	**Number of Studies**
	Self-efficacy theory	1

*Manchola-Gonzalez article did not report gender

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Table 5:

Study Quality Assessment

Criterion	Mean	SD
1. Was the research question/hypothesis clearly stated?	0.77	0.97
2. Were the inclusion and exclusion criteria clearly stated?	0.44	1.77
3. Was a clear aim for the study stated?	0.33	1.88
4. Were the methods of the study clearly described?	0.00	2.00
5. Were the main findings (results) of the study clearly described?	0.33	1.88
6. Were study limitations discussed?	0.50	1.66
7. Were physical activity recommendations in line with AICR/ACS guidelines?	1.05	0.88
8. Was a sample size justification via power analysis provided?	1.05	1.11
9. Are data analyses discussed?	0.44	1.77

Two authors of the review, S. Crowder and A. Buro independently scored each study based on these criteria, with disagreement resolved through discussion. Scores for each criterion range from 0 to 2, depending on whether the criterion was unmentioned or unmet (0), partially met (1), or completely met (2). The possible total study score ranges between 0 and 18. Higher total scores represent better quality. Study quality score assisted in the measure of the strength of study evidence; therefore, the purpose was not to exclude studies but rather to help the researchers gain a deeper understanding of each one.